Vidya Sudevan

Academic CV | November 2026

Postdoctoral Fellow, Computer and Information Engineering, Khalifa University, Abu Dhabi, UAE

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Research

Neuromorphic computing, spiking neural networks, underwater image enhancement, visual–inertial odometry, and semantic 3D underwater mapping.

Education

PhD in Engineering, specialization in Robotics, Khalifa University, 2021–2025. Dissertation: Dehazing Techniques for Navigation in Underwater Environments.

Master of Technology, Instrumentation and Control System, University of Calicut, 2014–2016.

Bachelor of Technology, Electrical and Electronics Engineering, University of Calicut, 2008–2012.

Research appointments

Postdoctoral Fellow, Khalifa University · July 2025–present

Doctoral Researcher, Khalifa University · August 2021–May 2025

Research Associate, Khalifa University · January–August 2021

Research Assistant, Khalifa University · November 2016–December 2020

Visiting Graduate Research Assistant, The Petroleum Institute · March–October 2016

Awarded research projects

ReefSight3D: Semantic 3D Software Stack for Underwater Ecological Monitoring and Marine Digital Twins. RI-IAG 2026. Lead proposal developer and project lead.

Visual-Haptics for Remote Robot Operation, Inspection & Physical Intervention. Khalifa University–Qatar University Joint Research Program 2025. Lead proposal developer and project lead.

Selected publications

Sudevan et al. *NeuroVIO: A Hybrid Neural Network Framework for Visual–Inertial Odometry Estimation.* IEEE Journal of Oceanic Engineering, 2026. DOI

Sudevan et al. *Spiking transformer with learnable threshold mechanism for underwater image dehazing to aid vision-based navigation.* Neuromorphic Computing and Engineering, 6, 014010, 2026. DOI

Teaching and supervision

Assigned subjects, 2021–2025: Artificial Intelligence, Machine Learning, Python Programming, and Mechatronics Labs. Co-supervision of two PhD candidates in Robotics.

Selected recognition

Second prize, 2nd Dubai World Challenge for Self-Driving Transport, Dubai RTA, 2021.